

X-ray Radiation Effects on SWCNT/PMMA Thin Film Nanocomposites

Guddi Suman , *Member, IEEE*, Merlyn Pulikkathara , and Richard Wilkins, *Member, IEEE*

Abstract—This paper studies the response of a radiation sensor that was developed from single-walled carbon nanotube (SWCNT) and poly (methyl methacrylate) (PMMA) nanocomposite thin films placed on an interdigitated electrode (IDE). We studied the effects of X-rays in real-time electrical resistance of the thin film nanocomposites. A 160 kV X-ray source irradiated the nanocomposite devices with ionizing X-ray radiation at different doses and dose rates, causing the decrease in resistance of thin film through the production of charge carriers. The embedded SWCNT network helped transport radiation-induced electrons to the electrodes. Pre-irradiation resistance can be obtained rapidly from heat treatment of irradiated nanocomposite devices. It is observed that the required heat treatment time for thin film device recovery increased almost linearly with applied radiation dose. The thin film devices were operated at dc voltage as low as 0.05 V for reliable measurement of the radiation-induced change in electrical resistance. The results of this study suggest that SWCNT/PMMA thin film devices could be used for sensing X-ray radiation, with significantly higher sensitivity than comparable devices made with multi-walled CNT (MWCNT) based nanocomposites in a previous study.

Index Terms—Single-walled carbon nanotube, poly (methyl methacrylate), interdigitated electrode, thin film, nanocomposite, N, N- dimethylformamide, X-ray, radiation detection.

I. INTRODUCTION

THE DETECTION of ionizing radiation is essential for biological and nuclear research, medicine, aerospace and defense applications. Various radiation detectors are used to measure and detect ionizing radiation. Many have several drawbacks: large volume, complicated instrumentation, incapability of producing a real-time signal, and low-temperature operation [1]. Thus, the search for real-time radiation detectors with excellent lifetime performance, environmental ruggedness, high accuracy, small form factor, and that are relatively light and economical remains a work in progress. Carbon nanotubes (CNTs)

are among the most explored nano-scale materials due to their unique electrical, thermal, and other physical properties [2]. Extraordinary properties such as the large length-to-diameter ratio, tunable bandgap, high electrical conductivity, high sensitivity to the surrounding environment, low cost, and functionalized ability make CNTs a promising nanomaterial for a variety of sensing applications [3]–[6], including ionizing radiation [1], [2]. According to previous research, CNTs are the most widely studied nanomaterials as gas sensors since they have chemically inert surfaces and a high surface area for adsorption sites for molecules [7]. For example, carbon nanotube-based sensors were used to detect SF₆ decomposed products and CNTs had shown to be sensitive to the presence of these gases [8]. Other gases such NO₂, H₂S, C₂H₅OH, and others have also been studied [9].

This paper studied the ionizing X-ray radiation effect on the electrical properties of SWCNT/PMMA thin film nanocomposites. Previous researchers investigated CNT thin film transistor (SWCNT-TFT), SWCNT thin film, and MWCNT/PMMA thin film composites as an X-ray radiation detector [10]–[12]. X-ray radiation effect on SWCNT was studied [13]. Numerical investigations of radiation-induced conductivity changes of carbon nanotube filled polymers were performed, using a 3D model to describe ionizing radiation-induced electrical conductivity of carbon nanotube-based nanocomposite [14].

The purpose of the continuing work described in this paper is to build a compact and durable radiation detector based on a rugged, thin film nanocomposite with a nanomaterial that provides real-time information. In previous work, MWCNT/PMMA thin film composites were studied to detect X-ray ionizing radiation [12]. Being simple, cheap, and easily processed, MWCNT nanocomposites could detect X-rays, but unfunctionalized MWCNT/PMMA thin film nanocomposite devices were limited in sensitivity, requiring functionalization to enhance X-ray response [16]. In the present work, SWCNT/PMMA thin film nanocomposites were irradiated with from a 160 kV 3000W X-ray source (X-RAD iR-160) to study the dose and dose rate effect on the resistance of SWCNT/PMMA thin film nanocomposites. The results showed a clear, repeatable, quasi-linear proportional response to applied radiation exposures of various total doses at fixed dose rates and of different dose rates to a specific total dose of X-rays. The observed results in this work showed similar response as reported in Ref. [12], with substantially increased sensitivity to X-ray exposure compared to unfunctionalized MWCNT/PMMA [12] and functionalized MWCNT/PMMA

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[16]. The results also showed that the SWCNT/PMMA thin film composite is about five times more sensitive to X-rays than similar unfunctionalized MWCNT/PMMA nanocomposites [12] and about two times more sensitive to X-rays than that of functionalized MWCNT/PMMA nanocomposites [16]. We further observed that post-irradiation heat treatment of the thin film nanocomposite devices could anneal the device to a condition close to pre-irradiation. To our knowledge, this is the first comprehensive study of X-ray radiation effects on thin film nanocomposites of SWCNT/PMMA.

II. MATERIALS AND METHODS

A. Materials to Make Thin Film Nanocomposite Device

We used the commercial unfunctionalized SWCNT as a nano-scale filler to prepare thin film nanocomposite devices (produced by a thermal chemical vapor deposition method greater than 95% purity) were obtained from Nanolab, Inc. with length 1–5 μm and diameter ~ 1.5 nm (D1.5L1-5-S, Nanolab, Inc.). Other materials used are poly (methyl methacrylate) (PMMA) as a polymer matrix (182230, Sigma Aldrich), N,N-dimethylformamide (DMF) as solvent (319937, Sigma Aldrich), and acetone for cleaning purposes obtained from Sigma Aldrich. The purpose of using DMF for the nanocomposite fabrication process is to give uniform dispersion of CNTs [17] in the solvent. PMMA is used during the fabrication of nanocomposite due to its good spinning qualities [18], high ruggedness [19], and good chemical resistance [20], solubility in N, N-dimethylformamide (DMF) [18]. The gold IDEs (#106a Electrodes, Case Western Reserve University) are deposited on top of a silicon substrate and are 0.5mm thick. The silicon substrate has dimensions of 8 mm $\times$ 8 mm, with a top layer of 300 nm thick silicon dioxide. The interdigitated electrode (IDE) and bonding pads (1.75 mm $\times$ 1.75 mm) consist of 200 nm thick gold with a 10 nm thick titanium adhesion layer. The electrodes run parallel to each other, with a line size spacing of 100 μm .

B. Preparation of SWCNT/PMMA Thin Film Nanocomposite Device

For the preparation of SWCNT/PMMA nanocomposite, 0.03g/cm³ concentration of PMMA in DMF was sonicated using a 750 W, 20 kHz probe sonicator (9810-24-750W, Ace Glass Inc.) for 15 minutes in an ice bath at 20% amplitude and 0.5 pulse rate. For the next step, 0.63 mg/cm³ concentration of SWCNT in DMF was added to the sonicated mixture of PMMA in DMF and sonicated in the ice bath again for 5 minutes at 20% amplitude and 0.5 pulse rate. The sonicated mixture of SWCNT/PMMA and DMF was deposited on the IDE by spin coating the mixture on the IDE at 1000 rpm with a ramp of 100 rpm/s for 30 seconds using a spin coater (model 200X, Brewer Science) to create a conducting SWCNT network atop the IDE gold electrode pattern, as shown in Fig. 1(a). The thin film composite cast on the IDE was cured in a vacuum oven at 175 $^{\circ}\text{C}$ for one hour and allowed to cool under a fume hood overnight to remove excess solvent. This process is also detailed in [Ref. 12]. Variable pressure scanning electron microscope (SEM, Hitachi SU3500) images of the SWCNT network are shown in Fig. 1(b) and of

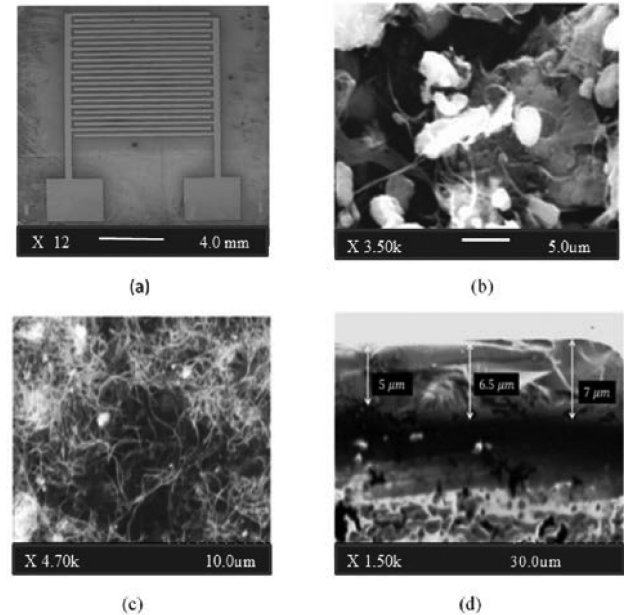


Fig. 1. (a) SEM image of IDE gold electrodes, (b) SEM image of unfunctionalized SWCNT powder, (c) SEM image of SWCNT/PMMA thin film composite, and (d) SEM image of SWCNT/PMMA thin film having a thickness in the range 5 μm to 7 μm .

the SWCNT/PMMA composite in Fig. 1(c). The nanocomposite thin film prepared by the above method had a thickness of about 5–7 μm as observed using a SEM, as shown in Fig. 1(d). The variable pressure SEM images are produced without any sample coating, and SEM voltages indicated in Fig. 1 are selected to optimize image quality. Fabricated thin film nanocomposites have 2.1% by weight of SWCNT/PMMA with resistances in the range 250–5000 Ω . While straightforward, this process can sometimes give varying results. For example, it has been observed that the degree of roughness of the tip used in the sonication process could degrade the performance of probe and resulted in poor dispersion of nanotubes in the polymer-solvent.

C. Experimental Setup

A 160 kV, 3000 W X-ray source (X-RAD iR-160) was used to perform the radiation exposure characterization of the thin film composites of nanomaterial [12]. The X-ray source has an X-ray tube with a tungsten target-anode at a focal spot of 7.5 mm. X-ray tube in which an energetic electron beam impinges a tungsten target, producing an X-ray spectrum that has sharply defined peaks at 55.55 keV and 66.66 keV with a low-lying broad bremsstrahlung background [12]. The resulting X-ray beam has a uniform circular cross-section that is about 15.24 cm in diameter at the surface of the tray in which the thin film device was positioned in the center of the tray. This beam diameter ensured that all irradiated thin film devices were uniformly exposed to the ionizing X-ray radiation in all experiments reported. The source to sample distance (SSD) was varied at 30 cm, 50 cm and 70 cm to study the X-ray radiation dose rate effects on the thin film nanocomposite devices. According to vendor's specification and calibration, the dose rate at 30 cm SSD is 5.92 krad /min. From the inverse square law, the calculated X-ray dose rates at 50 cm

SSD is 2.13 krad/min and at 70 cm SSD is 1.08 krad/min. Hence, the dose rates used are a function of calibrated measurements and the geometry of the X-ray machine. About 40 SWCNT/PMMA devices were used for this experiment.

Two experiments were studied in this paper. The first set of experiments was total dose effect on changes of resistance of the thin film device. Total dose is defined by varying the time the device was exposed from the source of X-rays. For example, when the device was placed at 30 cm from the center of the X-ray source this gives a dose rate of 5.92 krad/min. The device was irradiated to different X-ray total doses by varying sample exposure times. The second set of experiments was the dose rate effect on resistance of thin film devices; the dose rate is defined by varying the distance of the thin film device from the source of the X-rays. For example, a dose rate of 5.92 krad/min, 2.13 krad/min, and 1.08 krad/min is achieved by placing the thin film device at a distance 30 cm, 50 cm and 70 cm respectively from the center of the X-ray source. A Keysight B1500A semiconductor device parametric analyzer was used to carry out the electrical measurements on nanocomposite devices. The device was placed at the center of the sample tray inside the X-ray chamber and connected to the device analyzer using a low noise cable to perform the electrical measurement during irradiation. The Keysight B1500A was programmed to perform the electrical measurement for thin film device for pre-irradiation, during-irradiation, and post-irradiation. Except for results discussed in Section III C, all measurements were taken with an applied voltage of 0.5V and the current measured every second during an experiment. The experimental setup is described in further detail elsewhere Ref.12.

III. RESULTS

The X-ray radiation effect on the electrical resistance of SWCNT/PMMA thin film was obtained by measuring device resistance change during X-ray exposure. For a given measurement, the percentage change in resistance (S) of the SWCNT/PMMA thin film is defined by:

$$S = \left(\frac{R_f - R_i}{R_i} \right) \times 100\% \quad (1)$$

where R_f is the final resistance of SWCNT/PMMA thin film nanocomposite when X-ray irradiation ends, and R_i is the initial resistance of SWCNT/PMMA thin film before the X-ray irradiation. The calculated percentage change in nanocomposite resistance is the primary electrical characteristic used to show the effect of ionizing radiation on the nanocomposite films studied in this paper.

A. Total Dose Effect

In this section, the total dose effects were studied by recording real-time changes in the electrical resistance of the thin film device due to different total doses of X-ray radiation by varying the radiation exposure time at a constant dose rate. The devices used in the experiments had initial resistances in the range 2–3 k Ω . The device was placed at 30 cm SSD giving a dose rate of 5.92 krad/min. The device was irradiated to different X-ray

total doses of 40 krad, 100 krad, 200 krad, 500 krad, 750 krad and 1000 krad by varying exposure times 7 minutes, 17 minutes, 34 minutes, 85 minutes, 128 minutes, and 170 minutes respectively.

The electrical resistance of thin film device and an identical thin film control device (no X-ray exposure) were both recorded by the semiconductor device analyzer every second interval for pre-irradiation, during irradiation, and post-irradiation. The control device was not exposed to radiation. The pre-irradiation data typically recorded for 10 minutes and post-irradiation data for 80 minutes. Fig. 2(a) shows a typical data for an experiment. The resistance of the device substantially decreases continuously during irradiation for the different total doses, while the resistance of the control sample changes very little (typically less than 0.5%) during the experiment. Fig. 2(b) showed the change in resistance of SWCNT/PMMA nanocomposite for varying total doses at a constant dose rate 5.92 krad/min in the bar graph.

The percentage change from values obtained before irradiation is approximately directly proportional to applied radiation doses. Compared to the initial resistance before exposure, the thin film device that showed the lowest decrease in resistance at 7% was due to a total dose of 40 krad and the highest decrease in resistance at 58.1% was due to a total dose of 1000 krad. The post-irradiation data in Figs. 2(a) show that the device did not immediately recover the initial resistance after X-rays were turned off. Fig. 2(c) revealed the percentage change in resistance of thin film devices during X-ray exposures with varying X-ray doses at constant dose rate from the data in Fig. 2(a). The data points represent the mean percentage change in resistance of one device for multiple exposures (40 krad, 100 krad, 200 krad, 500 krad, 750 krad, and 1000 krad) with X-rays at dose rate 5.92 krad/min. The vertical bars represent standard deviation.

The experiment was repeated on four devices for multiple radiation exposures; all showed similar results. Fig. 2(c) showed that a linear equation could express the relationship between the change in resistance and X-ray radiation doses. The observed linear response is important for potential radiation detection applications of these nanocomposites.

B. Dose Rate Effect

The dose rate effect experiments were performed by varying the position of the device. For this set of experiments, we used the different dose rates based on the position of the device, to obtain a total dose of 100 krad for each dose rate. The dose rates that were studied were 5.92 krad/min, 2.13 krad/min, and 1.08 krad/min by placing the thin film device at 30 cm SSD, 50 cm SSD and 70 cm SSD respectively from the X-ray source origin in the X-ray chamber (Section II C). Fig. 3 shows changes in the electrical resistance of SWCNT/PMMA thin film device during exposure at the different dose rates to a total dose of 100 krad for each exposure. The times required to irradiate the devices at 100 krad for the different dose rates were calculated to be 92 minutes for 1.08 krad/min, 47 minutes for 2.13 krad/min and 17 minutes for 5.92 krad/min.

In Fig. 3, the SWCNT/PMMA resistance of device decreases over time during irradiation to X-rays of 100 krad total dose at

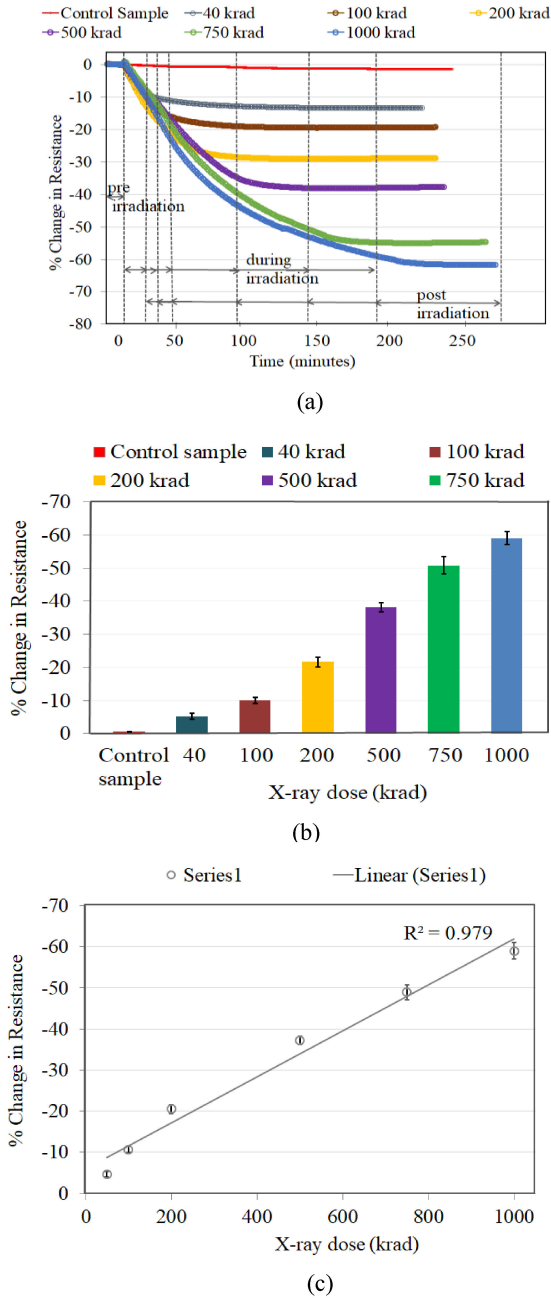


Fig. 2. (a) Change in resistance over time for SWCNT/PMMA nanocomposite for various total doses at a constant dose rate of 5.92 krad/min. (b) Change in resistance of SWCNT/PMMA nanocomposite for varying doses at a constant dose rate 5.92 krad/min in bar graph. (c) Change in resistance of SWCNT/PMMA nanocomposite for varying doses at a constant dose rate of 5.92 krad/min.

the different dose rates. The reduction in resistance of devices due to 100 krad X-ray exposure were 12.19% for 5.92 krad/min, 19.79% for 2.13 krad/min, and 29.92% for 1.08 krad/min. The decrease in resistance of the control device was very small, typically less than 0.5% during the experiment. The magnitude of the percentage change in resistance showed a roughly inverse proportionate response to exposed dose rates. For each different dose rate, the device had been subjected to heat treatment to reset the device to pre-irradiation resistance as described in

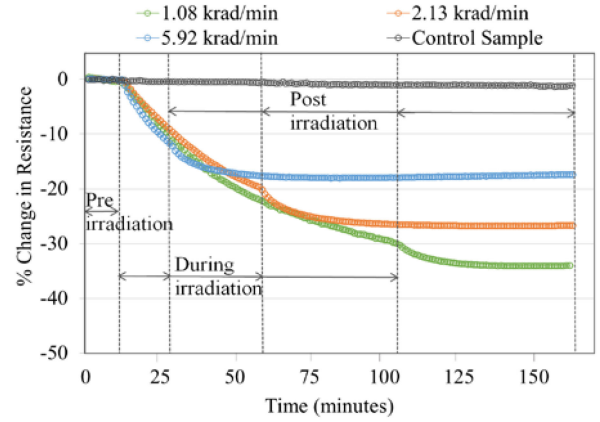


Fig. 3. Change in resistance of SWCNT/PMMA nanocomposite film with varying X-ray dose rates at constant X-ray dose at 100 krad.

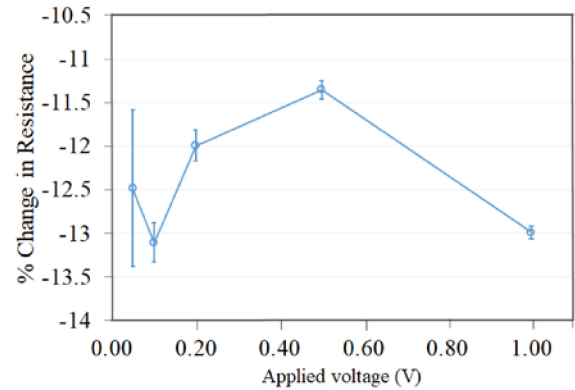


Fig. 4. Change in resistance of SWCNT/PMMA nanocomposite device due to 100 krad X-rays at 5.92 krad/min with various applied voltage, where vertical bars showing the standard deviation.

Section IV. The devices used for this experiment had initial resistances in the range 500–3000 Ω . This experiment was repeated on six devices for multiple exposures, which all showed similar results.

C. Effect of Lower Applied Voltage

In anticipation of using thin film devices as a radiation sensor, X-ray experiments were conducted on devices that were biased at voltages from 0.05 to 1.00 V. Fig. 4 shows percentage changes in resistance of one nanocomposite thin film device irradiated to 100 krad at 5.92 krad/min for different applied voltages starting from 0.05 V to 1.00 V. The data points in Fig. 4 indicated the mean percentage change in resistance of the thin film device, and vertical bars indicated standard deviation. It showed that the radiation-induced decrease in resistance of the thin film nanocomposite device from initial resistance before irradiation is approximately between 12% to 13% at all different voltages with the largest standard deviation at 0.05 V and the least standard deviation at 1.00 V. Fig. 4 also showed that there is a large error bar at 0.05 V, due to increased noise at lower applied voltage. The experiment was repeated on ten devices and had produced similar results. This result is essential for

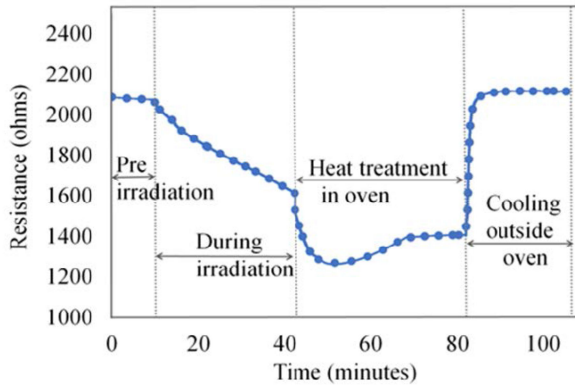


Fig. 5. Heat treatment recovery of SWCNT/PMMA device irradiated to 200 krad X-ray at 5.92 krad/min.

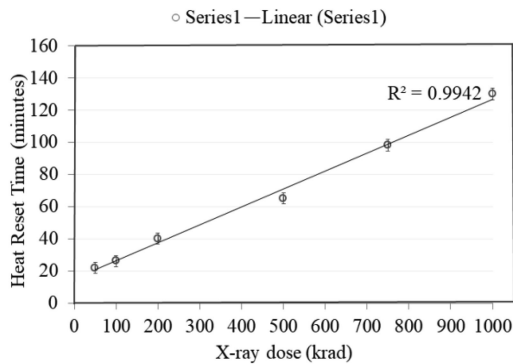


Fig. 6. Heat reset time variation with X-rays doses. The vertical bars represent standard deviation.

practical applications as it indicates the devices can operate at low voltages and hence with lower power consumption.

IV. HEAT TREATMENT AND RESET

The data in Figs. 2(a) and 3 indicate slow recovery of the radiation exposed nanocomposite devices. An irradiated device at 40 krad at ambient conditions slowly regains a resistance approximately the same as its pre-irradiation value after a few weeks, which was also observed in [Ref. 12]. Higher recovery times were observed for nanocomposite devices irradiated at higher X-rays total doses.

Heat treatment of the device based on nanotube and polymer reduces the naturally slow and long recovery time [12], [21]. Thus, the irradiated SWCNT/PMMA nanocomposite devices were subjected to heat treatment at 175 °C in an oven to accelerate the device recovery process after irradiation in this study. The radiated devices at 40 krad, 100 krad, 200 krad and 500 krad recovered their initial resistance prior to irradiation after heat treatment. Fig. 5 shows a typical heat treatment recovery of a device irradiated to 200 krad X-rays at 5.92 krad/min. The radiated device was heat treated for 45 minutes and then it took a further 8 minutes to recover the device. Fig. 6 shows the average heat reset times of the thin film device with applied X-ray doses 40 krad, 100 krad, 200 krad, 500 krad, 750 krad, and 1000 krad at 5.92 krad/min. The heat reset time of SWCNT/PMMA thin

film devices increases with increased X-ray total doses. The increase in X-ray radiation dose increased resistance change, requiring a longer heat treatment cycle as was seen also with the MWNT/PMMA, which observed in [Ref. 12]. In this work, the radiated device at 750 krad and 1000 krad did not fully recover the resistance to pre-irradiation value, which may be due to permanent damage of the thin film device due to a very high X-ray dose. Lower dosed, up to 500 krad samples recovered within about 100% of their pre-irradiated resistance, however, the higher dosed devices irradiated to 750 krad and 1000 krad recovered to about 90% of their pre-irradiated resistances.

The radiation exposure and heat treatment experiment was repeated for eight devices, and similar results were observed in all cases. The variation in the recovery time of devices was less than 4%. Fig. 6 shows a clear linear relationship with an R^2 value of 0.9942 between heat reset time and applied X-ray total doses.

V. DISCUSSION

The decrease in resistance of SWCNT/PMMA nanocomposite thin film devices during X-ray exposure was observed, as shown in Figs. 2(a) and 3. The X-rays cause an ionization effect through the production of charge carriers and increase the conductivity of the nanocomposite film, similar to that observed in carbon-fiber based devices [22]. The radiation generated charge carriers are transported to the electrodes by the network of nanotubes spanning between the electrodes [12]. The MWCNT network is essential for the charge transport; similar films made from carbon black do not show a significant response to X-ray radiation [12]. The decrease in resistance of SWCNT/PMMA nanocomposite thin film devices (as shown in Figs. 2(a) and (3)) is due to the X-ray irradiation is consistent with the result observed in SWCNT thin films device [23] and MWCNT/PMMA nanocomposite device [12].

Three feature effects were observed in Fig. 3: (1) the resistance of thin films in this work decreases during X-ray exposure, (2) the rate of change in resistance is directly proportional to dose rate, and (3) the magnitude of the decrease in resistance is inversely proportional to X-ray dose rate. These effects were also observed in the case of MWCNT nanocomposite device [12]. In this work, the resistance of nanocomposite film device continued decreasing very slowly even after the X-ray is turned off until it is stable, as shown in Fig. 2(a) and Fig. 3. This is likely due to the slow release of charges in the insulating polymer near the SWCNT-polymer interface. Fig. 2(a) and Fig. 3 also showed the decrease in resistance of less than 0.5% of the control sample during irradiation, which is very small compared to decrease in resistance of up to 58.1% due to 1000 krad X-ray irradiation. The reduction in resistance of the control device may be due to ambient humidity effects on SWCNTs [2]. Fig. 4 shows that the decrease in resistance of SWCNT/PMMA thin film device that were operated at the low voltages ranging from 0.05V and 1.00 V, was approximately 13% for all the applied voltages. However, the result showed more extensive scattered data at 0.05 V compared to a 1.00 V applied voltage, as shown in Fig. 4 is due to increased noise at a very low voltage signal.

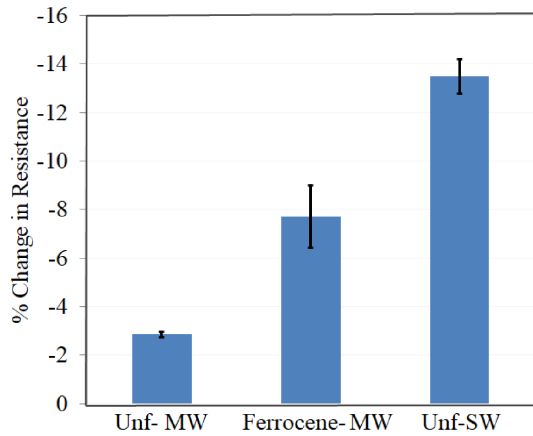


Fig. 7. Comparison of unfunctionalized SWCNT/PMMA responses with unfunctionalized MWCNT/PMMA [12] and ferrocene-doped MWCNT/PMMA [16] due to 100 krad X-rays at 5.92 krad/min.

In this work, the natural annealing of device at room temperature is a long process of about several weeks. Fast response due to high surface area of SWCNTs and low mobility of trapped holes in the oxide layer of IDE and polymer matrix probably contribute to the slow recovery of the resistance of thin film after irradiation [12], [25]. Thermal treatment is necessary to give sufficient energy to trapped holes to free them from such traps and help irradiated thin film nanocomposite to anneal in a much shorter time [12]. The higher the radiation dose the device is exposed to, the longer the device needs heat treatment during the annealing process, as shown in Fig. 6. This is consistent with the fact that higher doses would generate proportionally more trapped charges. In addition, the thin film devices did not recover completely their initial resistance after X-ray exposure to 750 krad and 1000 krad. This is likely because higher total X-ray doses cause destruction and disordering in polymeric structure [27]. X-ray exposure to nanotube reduces the sp^2 to sp^3 nature of the nanotube bond structure, as well as causing an increase in defect density, affecting electrical properties of nanotubes [13]. In addition, the heat treatment time for radiated nanocomposite device at 750 krad and 1000 krad are about 100 minutes and 130 minutes respectively. As the glass transition temperature of PMMA is 100.01 °C and melting point is 165 °C, heating SWCNT/PMMA nanocomposite at 175 °C for such longer times causes decomposition of composites which is associated with mass loss of PMMA in nanocomposites [26].

In this study, the X-ray radiation effect was studied in SWCNT/PMMA thin film nanocomposite device. A mixed type SWCNT (semiconducting and metallic) and PMMA was used to form thin film nanocomposite device. The reduction in resistance may be contributed by the ionization effect on both MWCNT and PMMA [16].

The data shown in Fig. 7 shows a comparison between thin film nanocomposite devices made from unfunctionalized SWCNT/PMMA and MWCNT/PMMA (both unfunctionalized and ferrocene-doped). There is a clear enhanced sensitivity (as characterized by the resistance change due to X-ray exposure) of unfunctionalized SWCNT/PMMA thin film com-

pared with functionalized MWCNT/PMMA as well as unfunctionalized MWCNT/PMMA to X-rays total dose of 100 krad at dose rate 5.92 krad/min. The overall superior sensitivity of the SWCNT/PMMA devices is also observed under other irradiation conditions. The decrease in resistance of unfunctionalized SWCNT/PMMA (Unf-SW) devices during X-ray exposure in this work was about five times higher than the MWCNT/PMMA (Unf-MW) device [12] and about two times higher than ferrocene-doped MWCNT/PMMA (ferrocene-MW) [16] due to 100 krad and 5.92 krad/min, as shown in Fig. 7. The ferrocene-MW data is shown because it had the highest sensitivity of MWCNT based devices previously studied [16]. The electrical properties of the CNTs network within the films are dependent on the chiralities; this effect is reduced when using MWCNT. Since the internal tubes are not connected to the outer tubes in MWCNTs, the contribution of the chirality of the inner tubes to their electrical properties are diminished and would have electronic distortion, in comparison to the singular chirality of SWCNT [24]. In addition, small diameter, larger aspect ratio, and larger surface area of SWCNTs compared with MWCNTs could be responsible for more charges produced and transported in SWCNT films compared with MWCNT films during radiation exposure. The morphological difference between SWCNTs and MWCNTs networks leads to more favorable conditions for forming crosslinks between SWCNTs than MWCNTs [2], providing more current paths causing more change in conductivity in SWCNT based devices during X-ray exposure compared with MWCNT based devices.

A 3D model to describe the electrical conductivity of CNT filled polymer polydimethylsiloxane (PDMS) during exposure to ionizing radiation was studied using the Monte Carlo method [14]. The researcher used this method to calculate CNT concentration, alignment's influence on electrical conductivity of carbon-filled polymer. A similar model cannot apply in this study due to several reasons: (1) during the fabrication of SWCNT/PMMA nanocomposite thin film, the sonicated composite was spin-cast on IDE to form thin film using a spin coater, which results in an indefinite number of nanotubes trapped in thin film, (2) SWCNTs used in this study is a mixed type containing semiconducting and metallic SWCNT and (3) SWCNT/PMMA is spin-coated on the interdigitated electrode, so there are a considerably large number of paths between individual SWCNTs, SWCNTs/PMMA, and SWCNTs/IDE finger surface created which increases the conductivity of the nanocomposite and lowering the overall resistance of the composite. Further experimental studying may help control the number of SWCNTs trapped in the desired region and regular alignment of all SWCNTs across the electrodes.

VI. CONCLUSION

This study shows that the percentage change in resistance of thin film nanocomposite devices based on SWCNT/PMMA shows a repeatable quasi-linear response to various applied total doses and different dose rates of ionizing radiation in the form of X-rays. The percentage change in resistance of thin film nanocomposites is directly proportional to total dose and

inversely proportional to the dose rate of X-rays which is consistent with the results observed in previous work [12]. In this work, the percentage of change in resistance of SWCNT/PMMA composite due to X-rays 100 krad at 1.08 krad/min and 5.92 krad/min are about five times and three times higher respectively than that observed in MWCNT/PMMA [12]. This demonstrates that composites made from SWCNT/PMMA are more sensitive to ionizing radiation than those made with MWCNT/PMMA [12]. The devices can be reset with thermal treatment. However, the thin film devices did not recover their initial resistance after 750 and 1000 krad X-rays exposure. The sensors could operate at low voltage signals with low power consumption, a favorable result for future operational applications. The results discussed here show that SWCNT/PMMA thin film devices effectively detect high dose rate X-rays ionizing radiation. Still, improvements for proper instrumentation of the device are necessary to isolate the device from environmental humidity and other environment factors that may change device resistance. The dose rates used in this experiment are comparatively much higher than those expected in the natural environment. Future work will include studying the lower dose rates radiation and gamma radiation effect on nanocomposite devices, which is essential to evaluate the performance of thin film composites for many applications.

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REFERENCES

- [1] J. M. Lobe, and T. M. Swager, "Radiation detection: Resistivity responses in functional poly (Olefin sulfone)/carbon nanotube composites," *Angew. Chem. Int. Ed.*, vol. 49, pp. 95–98, Jan. 2010, doi: 10.1002/anie.200904936.
- [2] A. Nelson, "Carbon nanotube based dosimetry of neutron and gamma radiation," Ph.D. dissertation, Eng. Mechanics, Virginia Polytech. Ins. and State Univ., Blacksburg, VA, Apr. 29, 2016. Accessed on: Oct. 22, 2017. [Online]. Available: https://vtechworks.lib.vt.edu/bitstream/handle/10919/79732/Nelson_AJ_D_2016.pdf?sequence=1&isAllowed=y
- [3] C. Li, E. Thostenson, and T. Chou, "Sensors and actuators based on carbon nanotubes and their composites: A review," *Compos. Sci. Technol.*, vol. 68, no. 6, pp. 1227–1249, May 2008.
- [4] N. Sinha, J. Ma, and J. T. W. Yeow, "Carbon nanotube-based sensors," *J. Nanosci. Nanotechnol.*, vol. 6, no. 3, pp. 573–590, Feb. 2006.
- [5] B. Mahar, C. Laslau, R. Yip, and Y. Sun, "Development of carbon nanotube-based sensors- A review," *IEEE Sensors J.*, vol. 7, no. 2, pp. 266–284, Feb. 2007.
- [6] D. Ziegler, E. Bekyarova, A. Marchisio, J. M. Tulliani, and K. Naishadham, "Highly selective ozone sensors based on functionalized carbon nanotubes," in *Proc. IEEE SENSORS*, 2018, pp. 1–4.
- [7] R. Xuemei, C. Chen, M. Nagatsu, and X. Wang, "Carbon nanotube as adsorbents in environmental pollution management: A review," *Chem. Eng. J.*, vol. 170, no. 2-3, pp. 395–410, Jun. 2011.
- [8] X. Zhang, H. Cui, Y. Gui, and J. Tang, "Mechanism and application of carbon nanotubes sensors in SF6 decomposed production detection: A review," *Nanoscale Res. Lett.*, vol. 12, no. 177, pp. 1–12, Mar. 2017.
- [9] T. Han, A. Nag, S. C. Mukhopadhyay, and Y. Xu, "Carbon nanotubes and its gas-sensing applications: A review," *Sensors Actuators A: Phys.*, vol. 291, pp. 107–143, Jun. 2019, [Online]. Available: <https://doi.org/10.1016/j.sna.2019.03.053>
- [10] X. W. Tang, *et al.*, "Measurement of ionizing radiation using carbon nanotube field effect transistor," *Phys. Med. Biol.*, vol. 50, no. 3, pp. N23–N31, Feb. 2005.
- [11] Q. Kang, and J. T. W. Yeow, "Electrical resistance response evaluation of semiconducting single-walled carbon nanotube film for X-ray sensing," in *Proc. IEEE 11th Int. Conf. Nanotechnol.*, 2011, pp. 387–391.
- [12] S. Kyatsandra, and R. Wilkins, "Total ionizing dose X-ray radiation effects on MWCNT/PMMA thin film composites," *IEEE Trans. Nanotechnol.*, vol. 14, no. 1, pp. 152–158, Jan. 2015.
- [13] N. Bardi, I. Jurewicz, A. K. King, M. A. Alkhorayef, D. Bradley, and A. B. Dalton, "X-ray irradiation-induced structural changes on Single-wall carbon nanotubes," *Radiat. Phys. Chem.*, vol. 140, pp. 34–37, Nov. 2017, [Online]. Available: <http://dx.doi.org/10.1016/j.radphyschem.2017.03.008>
- [14] F. Liu, Y. Sun, W. Sun, Z. Sun, and J. T. W. Yeow, "Numerical and experimental study of radiation-induced conductivity change of carbon nanotube filled polymers," *Nanotechnology*, vol. 28, Jun. 2017, Art. no. 255501.
- [15] E. X. Zhang, *et al.*, "Low-energy X-ray and ozone-exposure induced defect formation in graphene materials and devices," *IEEE Trans. Nucl. Sci.*, vol. 58, no. 6, pp. 2961–2967, Dec. 2011.
- [16] S. Kyatsandra, M. Pulikkathara, and R. Wilkins, "X-ray radiation effects on thin film nanocomposites of functionalized and copper coated multi-walled carbon nanotube and poly(methyl methacrylate)," *Surf. Interfaces*, vol. 17, Jul. 2019, Art. no. 100362, [Online]. Available: <https://doi.org/10.1016/j.surf.2019.100362>
- [17] H. Ham, Y. Choi, and I. Chung, "An explanation of dispersion states of single-walled carbon nanotubes in solvents and aqueous surfactant solutions using solubility parameters," *J. Colloid Interface Sci.*, vol. 286, no. 1, pp. 216–223, Jun. 2005.
- [18] F. Du, J. E. Fischer, and K. I. Winey, "Coagulation method for preparing single-walled carbon nanotube/poly (methyl methacrylate) composites and their modulus, electrical conductivity, and thermal stability," *J. Polym. Sci. B, Polym. Phys.*, vol. 41, no. 24, pp. 3333–3338, Dec. 2003.
- [19] J. E. Mark, "Poly (methyl methacrylate)," in *Polymer Data Handbook*, 1st ed., L. Lu and A. G. Mikos, Eds., New York, NY, USA: Oxford Univ. Press, pp. 655–657, 1999. [Online]. Available: http://www.toarplast.co.il/linksPage/Hand_Book.pdf
- [20] W. F. Smith, and J. Hashemi, "Polymeric materials," in *Foundations of Materials Science and Engineering*, 4th ed., W. D. Callister and D. G. Rethwisch, Eds., New York, NY, USA: McGraw-Hill, 2005, p. 510.
- [21] W. D. Zhang, and W.-H. Zhang, "Carbon nanotubes as active components for gas sensors," *J. Sens.*, vol. 2009, pp. 1–16, Apr. 2009, Art. no. 160698, doi: 10.1155/2009/160698.
- [22] J. Ma, J. Yeow, J. Chow, and R. B. Barnett, "A carbon fiber-based radiation sensor for dosimetric measurement in radiotherapy," *Carbon*, vol. 46, no. 14, pp. 1869–1873, Nov. 2008.
- [23] J. Ma, J. Yeow, J. Chow, and R. B. Barnett, "A transparent carbon nanotube sensor for radiation detection," in *Proc. IEEE Sens.*, 2008, pp. 1394–1397.
- [24] A. Lekawa-Raus, J. Patmore, L. Kurzepa, J. Bulmer, and K. Koziol, "Electrical properties of carbon nanotube-based fibers and their future use in electrical wiring," *Adv. Funct. Mater.*, vol. 24, pp. 3661–3682, 2014, [Online]. Available: <https://doi.org/10.1016/j.sna.2019.03.053>
- [25] L. Camilli, and M. Passacantando, "Advances on sensors based on carbon nanotubes," *Chemosensors*, vol. 6, no. 4, p. 62, Dec. 2018.
- [26] S. Iqbal, M. S. Rafique, S. Anjum, A. Hayat, and N. Iqbal, "Impact of X-ray irradiation on PMMA thin films," *Appl. Surf. Sci.*, vol. 259, pp. 853–860, Aug. 2012, [Online]. Available: <http://dx.doi.org/10.1016/j.apsusc.2012.07.146>
- [27] P. Kalakonda, and S. Banne, "Thermomechanical properties of PMMA and modified SWCNT composites," *Nanotechnol., Sci. Appl.*, vol. 10, pp. 45–52, Feb. 2017, [Online]. Available: <http://dx.doi.org/10.2147/NSA.S123734>



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